Evaluating Language Models

Perplexity · BLEU · ROUGE · BERTScore · LLM-as-Judge · Benchmarks

Outline

The evaluation problem

Perplexity

n-gram overlap metrics

Ranking & retrieval metrics

Model-based evaluation

Benchmarks & wrap-up

Why evaluation is hard

Prompt: "Summarize the key findings of the study on climate change."

Output A:
"The study found that global temperatures rose 1.2°C since 1900, with accelerating ice loss in the Arctic."

Output B:
"The research demonstrates significant atmospheric changes leading to enhanced precipitation patterns globally."

Multiple valid outputs
Many correct ways to say the same thing

Meaning vs. form
Surface similarity $\neq$ semantic similarity

Task-dependent
Translation, summary, chat need different metrics

Intrinsic vs. extrinsic evaluation

Intrinsic

Measures the model *itself*

Perplexity
Cross-entropy loss
Bits-per-character

"How well does the model predict?"

Low PP $\neq$ good output

Extrinsic

Measures down-stream *task* quality

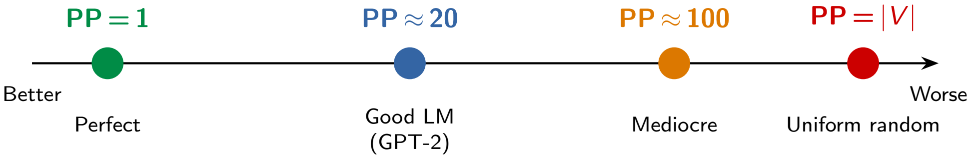
BLEU (translation)
ROUGE (summarization)
Accuracy / F1

"How useful is the output?"

Perplexity

$$\mathrm{PP}(W) \;=\; \exp\left(-\frac{1}{N}\sum_{i=1}^{N}\log P(w_i \mid w_{<i})\right)$$

Intuition: On average, the model is as uncertain as choosing uniformly among **PP** tokens at each step — the *effective branching factor* (out of the whole vocabulary).



Same thing: $\mathrm{PP} = \exp(H)$, where $H = -\frac{1}{N}\sum \log P(w_i \mid w_{<i})$ is the *cross-entropy* (average negative log-likelihood).

Perplexity — worked example

Sequence "the cat sat" ($N = 3$ tokens), natural log

token	$P(w_i \mid \textbf{context})$	$-\log P$
the	0.5	0.69
cat	0.4	0.92
sat	0.2	1.61
sum		**3.22**

Average negative log-likelihood: $\frac{1}{N}\sum_i(-\log P) = \frac{3.22}{3} = 1.073$

$$\mathrm{PP} = \exp(1.073) \approx 2.9$$

Read it as: at each step the model is about as unsure as a fair ~**3-sided die**.

Confident ($P \approx 0.9$ each): $\mathrm{PP} \approx 1.1$. Uniform over vocab: $\mathrm{PP} = |V|$.

Perplexity — visual intuition

Predicting the next token after: "The cat sat on the ______"

Lower perplexity = model concentrates probability on the right tokens (bars illustrative, not exact)

Perplexity — caveats

1. Tokenizer-dependent

Different BPE vocabularies produce different N — PP scores across models with different tokenizers are *not* comparable.

2. Not a quality metric

Low PP means good prediction, not good *content*. A model can confidently produce fluent nonsense.

3. Memorization

A model that memorizes training data has very low PP on that data but fails on new text. Check on *held-out* data.

4. LM-only

Perplexity only applies to generative (autoregressive) models. For classification, QA, etc. use task-specific metrics.

Takeaway: PP is great for comparing LMs *on the same data and tokenizer*, but don't over-interpret it.

BLEU — Bilingual Evaluation Understudy

Papineni et al., 2002 — designed for machine translation

Reference:	The	cat	is	on	the	mat
Candidate:	The	cat	sat	on	the	mat

match

no match

$$p_n = \frac{\text{matched } n\text{-grams in candidate}}{\text{total } n\text{-grams in candidate}}$$

Unigram precision: $p_1 = 5/6 \approx 0.83$ Bigram precision: $p_2 = 3/5 = 0.60$

(raw counts; denominators count *candidate n*-grams — the "clipping" fix comes next slide)

Core idea: count how many *n*-grams in the candidate also appear in the reference

BLEU — modified precision & brevity penalty

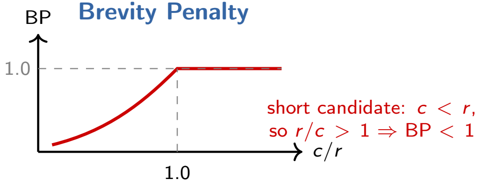
Problem: "the the the the the" has 100% unigram precision against any sentence containing "the"!

$$\text{BLEU} = \underbrace{\text{BP}}_{\text{brevity penalty}} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

$$\text{BP} = \min\left(1, \exp(1 - r/c)\right)$$

r = reference length, c = candidate length, $N = 4$, $w_n = \frac{1}{4}$

Brevity Penalty



BLEU-4 is standard:

geometric mean of p_1, p_2, p_3, p_4

Higher $n \Rightarrow$ stricter fluency check

BLEU — worked example

Reference: "The cat is sitting on the mat" $r = 7$

Candidate: "The the cat mat" $c = 4$

Clipped precision by n-gram order

n	**Candidate n-grams**	**Clipped matches**	p_n
1	4	3	3/4
2	3	1	1/3
3	2	0	0/2
4	1	0	0/1

$$\mathrm{BP} = \exp(1 - 7/4) = \exp(-0.75) \approx 0.47$$

$p_3 = p_4 = 0 \;\Rightarrow\;$ geometric mean $= 0 \;\Rightarrow\;$ raw BLEU $= 0$.

A single zero p_n wipes out the whole score, so real BLEU adds *smoothing* (a tiny ε for zero counts). Either way, this short candidate scores very low.

ROUGE — Recall-Oriented Understudy for Gisting Evaluation

BLEU asks: "How many candidate *n*-grams appear in the reference?" (**precision**)

ROUGE asks: "How many *reference* *n*-grams appear in the candidate?" (**recall**)

$$\text{ROUGE-}N = \frac{\text{matched } n\text{-grams}}{\text{total } n\text{-grams in } \textbf{reference}}$$

ROUGE-1

Unigram recall

Content coverage

ROUGE-2

Bigram recall

Fluency + content

ROUGE-L

Longest common subsequence

LCS length / ref length

A *subsequence* keeps word order but need not be contiguous — LCS of "the big cat sat" and "the cat sat" is "the cat sat" (length 3). ROUGE-1/2/L are usually all reported, often as F1.

BLEU vs. ROUGE

BLEU

Precision-based

"What fraction of candidate n-grams are correct?"

Best for: **Translation**

ROUGE

Recall-based

"What fraction of reference content is captured?"

Best for: **Summarization**

	BLEU	**ROUGE**
Focus	Precision	Recall (+ F1)
Penalizes short output	Yes (BP)	No
Penalizes missing content	No	Yes
Typical n	1–4 (geometric mean)	1, 2, or L
Range	0–1 (often $\times$100)	0–1

Limitations of *n*-gram metrics

Reference: "The movie was really excellent"

Candidate A:
"The film was truly great"
Good paraphrase, low BLEU

Candidate B:
"The movie was really bad"
Wrong meaning, high BLEU

Synonyms ignored
"great" $\neq$ "excellent" even though they mean the same

Semantics missed
"really bad" vs. "really excellent" share most *n*-grams

Multiple references needed
One reference can't capture all valid translations / summaries

Open-ended generation
No reference exists for creative writing, chat, etc.

Modern alternatives: **BERTScore** (embedding similarity), **METEOR** (synonyms + stemming), **COMET** (learned metric), **LLM-as-Judge**

Ranking metrics: is the relevant stuff near the top?

In search / retrieval / RAG the system returns a **ranked list**; we score *where* the relevant items land. Say 3 relevant docs exist and the top 5 returned are:

×	✓	✓	×	✓
rank 1	rank 2	rank 3	rank 4	rank 5

Precision@k

rel in top-k / k

$P@3 = 2/3$

Recall@k

rel in top-k / total rel

$R@3 = 2/3$

MRR

mean of

1/(rank of 1st rel)

$RR = 1/2$

Precision@k / **Recall@k** ignore order *within* the top-k.
MRR rewards getting *one* relevant hit high (great for QA with a single answer). For full-ordering quality, next slide.

Rank-aware: MAP and nDCG

Same ranked list (relevant at ranks 2, 3, 5; 3 relevant total). These reward putting relevant items *higher*.

Average Precision (AP)

average of P@k at each relevant rank:

$\mathrm{AP} = \frac{1}{3}\left(\frac{1}{2} + \frac{2}{3} + \frac{3}{5}\right) \approx 0.59$

MAP = mean AP over all queries

nDCG

$\mathrm{DCG} = \sum_i \frac{\mathrm{rel}_i}{\log_2(i+1)}$, normalized by the ideal ranking:

$$\mathrm{nDCG} = \frac{1.52}{2.13} \approx 0.71$$

handles **graded** relevance, not just yes/no

nDCG is the standard for graded relevance — exactly what the RAG deck reports (`nDCG@10`). **MAP** is the standard binary-relevance summary. Both reward a good *ordering*, not just a good set.

BERTScore — greedy token matching

Zhang et al., ICLR 2020 — replace *n*-gram matching with contextual-embedding similarity

Idea: embed each token (e.g. RoBERTa), then match every token to its *most similar* token in the other sentence by cosine similarity.

Candidate

Reference		a	cat	sits
	the	0.5	0.1	0.1
	cat	0.1	1.0	0.2
	sat	0.1	0.2	0.8

Green = each token's best match. Recall $= \frac{0.5+1.0+0.8}{3} = 0.77$ (rows), Precision $= 0.77$ (cols) — *equal here by coincidence; usually $P \neq R$.*

BERTScore — P, R, F1 and trade-offs

From the greedy matches: P averages over *candidate* tokens, R over *reference* tokens, $F_1 = 2\,\frac{P \cdot R}{P + R}$ (usually reported).

Strengths

- Captures synonyms & paraphrases
- Correlates with humans better than BLEU/ROUGE
- Works across many languages
- No training — off-the-shelf LM

Limitations

- Score depends on the embedding model
- Slower than BLEU/ROUGE (needs a GPU)
- Still reference-based: needs ground truth
- Sentence-level, not discourse-level

Cousins: METEOR (synonyms + stemming); **COMET** / **BLEURT** are metrics *trained* on human ratings, pushing correlation with humans even higher.

LLM-as-Judge — pointwise vs pairwise

Zheng et al., 2023 — a dominant paradigm for open-ended generation (no reference needed)

Core idea: use a strong LLM (e.g. GPT-4) to score or rank other models' outputs.

Pointwise (absolute)

Judge scores one answer, e.g. 1–10 on helpfulness / accuracy.

e.g. **MT-Bench**: 80 multi-turn Qs, 8 categories

Scores drift; hard to calibrate

Pairwise (relative)

Judge picks the better of A vs B; aggregate many duels into **Elo** ratings.

e.g. **Chatbot Arena**: 1M+ blind human votes

Easier call, more reliable

Why pairwise wins: "which is better?" is an easier, more consistent judgment than "rate this 7/10." Elo turns pairwise wins into a global ranking (like chess).

LLM-as-Judge — biases & why it dominates

Known biases

- **Position**: favors the answer shown first
- **Verbosity**: longer looks "better"
- **Self-enhancement**: prefers its own family
- **Style over substance**: fluent > correct

Why it dominates

- Scales to any open-ended task
- No reference texts required
- High correlation with human judgment
- 10–100× cheaper than human eval

Mitigations: swap A/B order and average (cancels position bias); control for length; use a *jury* of several judges; keep humans in the loop for high-stakes calls.

Benchmarks — evaluating holistically

Instead of one metric, aggregate many tasks to measure general capability

GLUE 2018	SuperGLUE 2019	MMLU 2021	Big-Bench 2022	Chatbot Arena 2023
9 NLU tasks: sentiment, NLI, paraphrase, QA	8 harder NLU tasks: coreference, causal reasoning	57 subjects: STEM, humanities, social sciences	200+ diverse tasks: reasoning, math, knowledge	Human pairwise preference (Elo ratings)

Key idea: single metrics are fragile — bench-marks aggregate many sub-tasks for a holistic view

Benchmark saturation: GLUE was "solved" (models passed the human baseline) within a year — SuperGLUE soon after. The bar keeps rising.

Key benchmarks — knowledge & language

Benchmark	What it tests	Format	Notable for
MMLU	Knowledge (57 subjects)	Multiple choice	Breadth of knowledge
MMLU-Pro	Knowledge (harder)	Multiple choice (10-way)	Reasoning-heavier MMLU
GPQA	Graduate-level science	Multiple choice	"Google-proof" PhD questions
TruthfulQA	Factual accuracy	QA	Hallucination resistance
HellaSwag	Commonsense	Sentence completion	Everyday physical reasoning
Winogrande	Coreference	Pronoun resolution	Linguistic understanding

Mostly *multiple choice* $\Rightarrow$ trivial to score automatically — but partly guessable (4 options = 25% for free) and easy to contaminate.

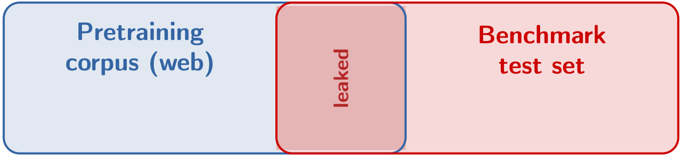
Key benchmarks — reasoning, code & agents

Benchmark	What it tests	Format	Notable for
GSM8K	Grade-school math	Word problems	Step-by-step reasoning
ARC	Science questions	Multiple choice	Grade-school science
HumanEval	Code generation	Function + hidden tests	Coding ability (pass@k)
SWE-Bench	Coding agents	Real GitHub issues	End-to-end software eng.
LiveBench	Mixed, refreshed monthly	Various	Contamination-resistant

Aggregate leaderboards (e.g. **Open LLM Leaderboard**, **HELM**) rank models at a glance — but no single number tells the full story.

Benchmark contamination

The leaderboard's dirty secret: test items leak into pretraining data



Symptom

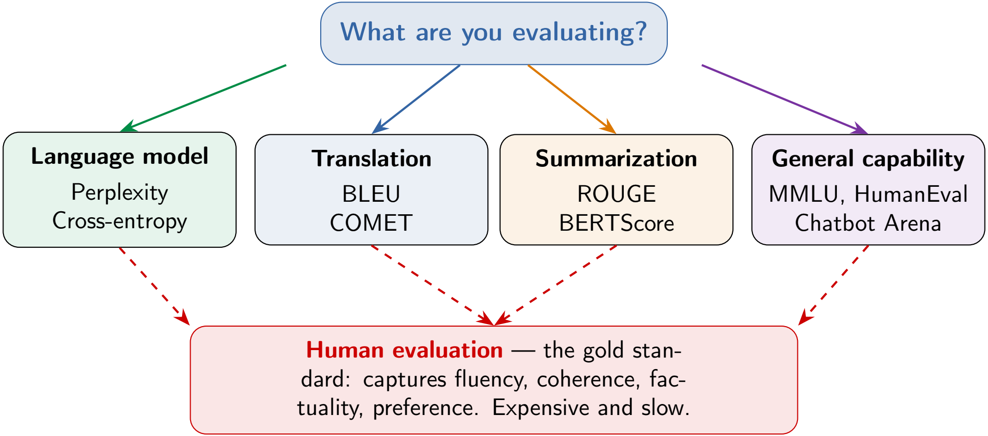
Answers are memorized ⇒ inflated scores. The gap shows on *rephrased* or *newer* items (e.g. GSM8K vs a freshly written GSM1k).

Defenses

Private/held-out sets; *canary* strings (a unique marker phrase — if a model can recite it, it trained on the test set); monthly refresh (LiveBench); re-collected variants; overlap checks.

Rule of thumb: a score is only meaningful if the model has not already seen the answers.

The evaluation landscape



No single metric captures everything — use the right tool for the right task

Further reading

Classic Metrics

- Papineni et al. (2002), "BLEU: A Method for Automatic Evaluation of Machine Translation"
- Lin (2004), "ROUGE: A Package for Automatic Evaluation of Summaries"
- Banerjee & Lavie (2005), "METEOR: An Automatic Metric for MT Evaluation"

Modern Evaluation

- Zhang et al. (2020), "BERTScore: Evaluating Text Generation with BERT"
- Zheng et al. (2024), "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena"

Benchmarks & Leaderboards

- Srivastava et al. (2023), "Beyond the Imitation Game" (BIG-Bench)
- Hendrycks et al. (2021), "Measuring Massive Multitask Language Understanding" (MMLU)

Questions?

Next: Early Notable Models — GPT, BERT, T5